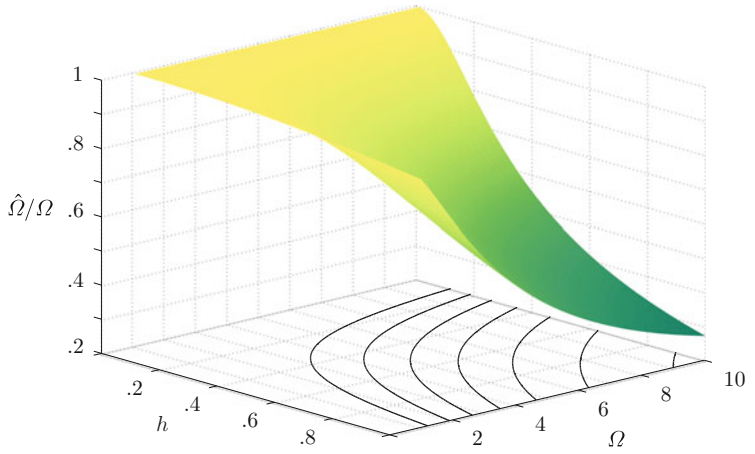


Fig. 4.2 The frequency ratio $\hat{\Omega}/\Omega$ for the Implicit Midpoint method. A frequency ratio of 1 indicates that the phase of the harmonic oscillator is accurate

boundedness of the timestepping iteration, and then only at the point of linearization. Other issues are often in play.

If the Implicit Midpoint method is viewed as the exact solution of a perturbed harmonic oscillator, then the frequency of this oscillator is

$$\hat{\Omega} = \frac{1}{ih} \ln \frac{1 + ih\Omega/2}{1 - ih\Omega/2}.$$

In the diagram below (Fig. 4.2), we plot the ratio $\hat{\Omega}/\Omega$ against h and Ω. If this ratio is 1, the frequency is accurately represented. The figure shows that the frequency is only well approximated in case $h\Omega$ is sufficiently small. For $h\Omega \gg 1$ the frequency is highly modified. From this we might infer that a molecular dynamics simulation performed using the Implicit Midpoint method is only likely to be accurate in terms of resolution of temporally correlated quantities in case $h\Omega$ is relatively small.

The inaccuracy of phase contributions computed using an implicit method with large stepsize (much larger than the Verlet stability threshold, i.e. $h \gg 2/\Omega$) may, in complicated systems, have additional consequences for error growth. For example, consider a large harmonic system with slow and fast dynamical components. The frequency ratio shown in Fig. 4.2 suggests that, when using larger timesteps, the most severe errors will be introduced in approximating the fast variables. The modification of frequencies will mean that these components will resonate (in a nonphysical way) with the low frequency components, thus we would anticipate a rapid propagation of errors from fast components into slow variables (in practice an error in the bond stretch may lead, on a rapid timescale, to propagation into a dihedral bend). We refer the interested reader to more in-depth treatments of these issues [17, 152] for more details.